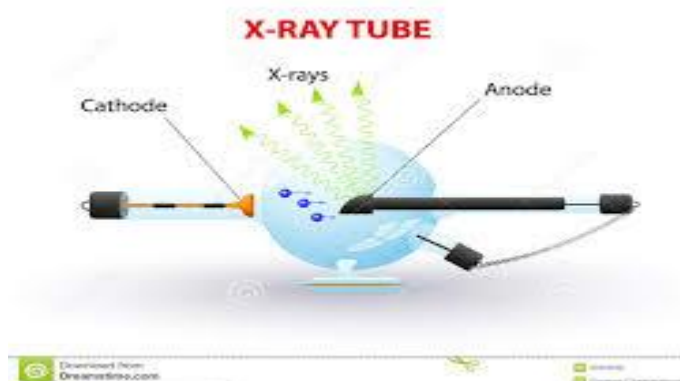




Title of Lecture: X-Ray Lab. I

Hawler Medical University
College of Medicine
Medical physics Units
Assisst.Prof. Dr.Ronak .T.Ali





Content:

1. Background information about x-rays.
2. How to minimize your radiation exposure
3. Applications of x-rays in medicine.



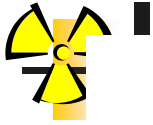


What are X-rays?

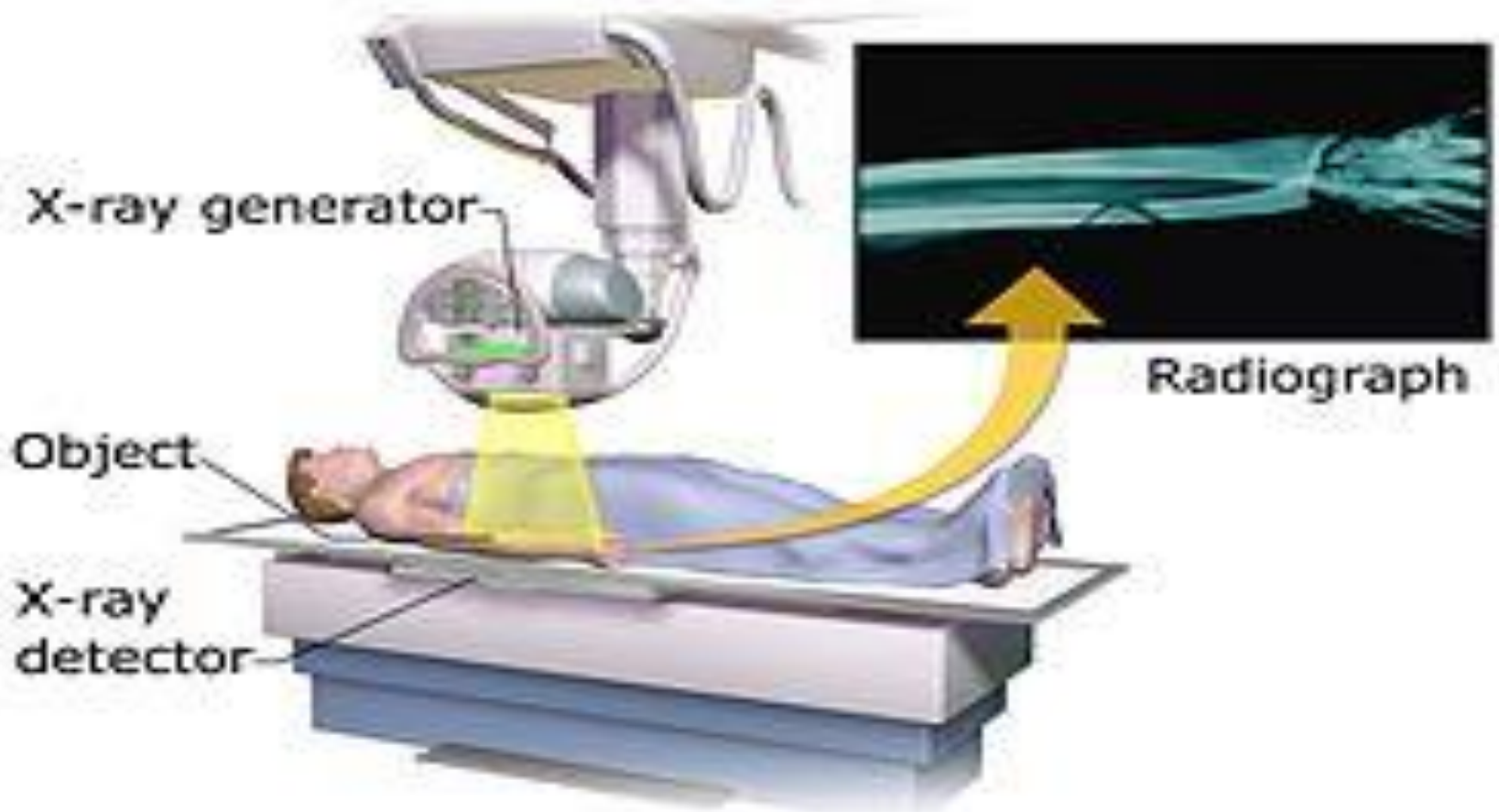
- X rays are a form of electromagnetic radiation which arises as electrons are deflected from their original paths or inner orbital electrons change their orbital levels around the atomic nucleus.
- X rays, like gamma rays, are capable of traveling long distances through air and most other materials.
- X and gamma rays differ primarily in their origin: x rays originate in the electronic shell, gamma rays originate in the nucleus.

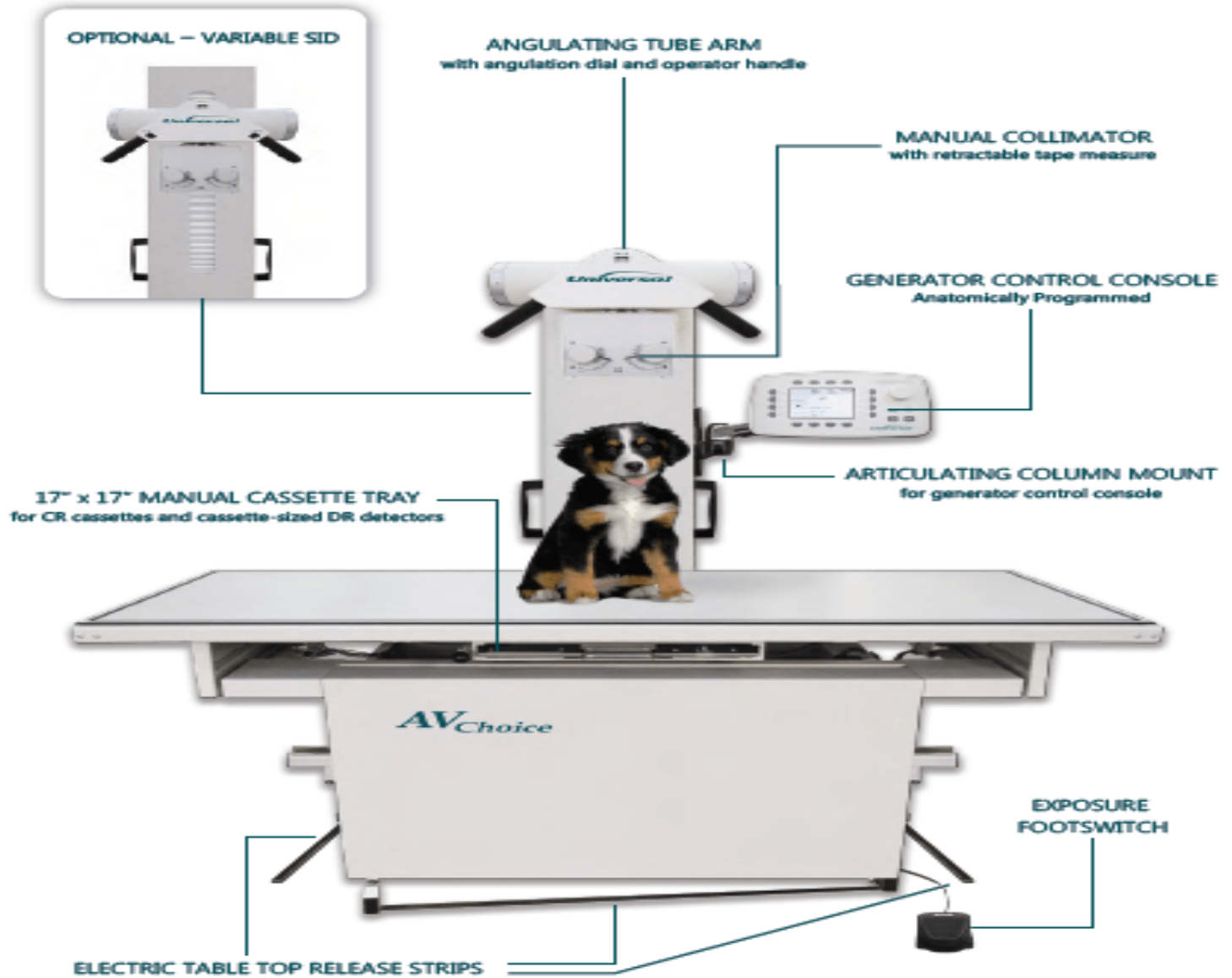


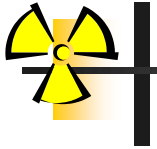
Early X-Ray Tube (1899): This tube is a specimen of the first type of gas x-ray tube to incorporate a water-cooled anode. The hollow anode was supplied with water by gravity feed from a supply held in the side bulb. This type of tube was introduced by Mueller about 1899.



Projectional radiography







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HAZARDS OF ANALYTICAL X-RAY EQUIPMENT

1. **The primary beam:** The primary beam is most hazardous because of the extremely high exposure rates. Exposure rates of 4×10^5 R/min at the port have been reported for ordinary diffraction tubes.
2. **Leakage or scatter of the primary beam through cracks in ill fitting or defective equipment:** The leakage or scatter of the primary beam through apertures in ill fitting or defective equipment can produce very high intensity beams of possibly small and irregular cross section.
3. **Penetration of the primary beam through the tube housing, shutters or diffraction apparatus:** The hazard resulting from penetration of the useful beam through shutters or the x-ray tube housing is slight in well designed equipment. Adequate shielding is easily attained at the energies commonly used for diffraction and florescence analysis.
4. **Diffracted rays:** Diffracted beams also tend to be small and irregular in shape. They may be directed at almost any angle with respect to the main beam, and occasionally involve exposure rates of the order of 80 R/h for short periods.



Diagnostic X-rays

Two main types of diagnostic X-ray devices:

Radiograph – a picture with film or image is sent direct to computer screen

Fluoroscopic – a real time “moving” inspection on inside functions

Diagnostic radiology is the branch of medicine that has traditionally been known for taking and reading X-rays. Like every other field of medicine, technology has radically changed this specialty forever. Diagnostic radiology is the nucleus of almost every physician’s diagnosis. Being able to detect disease sooner and pinpoint its location more accurately is a huge factor in stopping disease in its tracks.

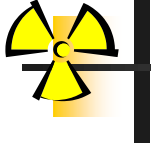


X-Ray Effects

The effects of x-ray exposure depends upon:

- ❑ **Duration - How fast the dose is delivered.**
- ❑ **Energy - How much energy was in the x-ray**
 - ❑ **Low Energy (<50 KeV) - damage only to skin or outer part of body**
 - ❑ **High Energy - damage to internal organs**
- ❑ **Total Dose - The magnitude of the dose**

How to minimize your radiation exposure



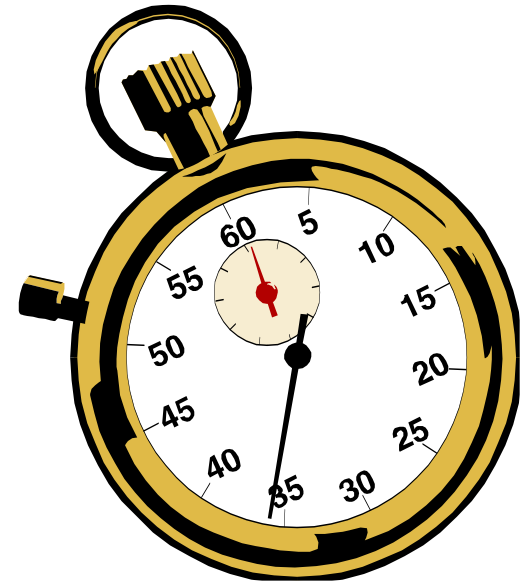
- Radiation use is very prevalent throughout hospitals
- Many steps have been taken to keep exposures to personnel at a minimum
- You are required to understand the nature of this use, how to keep exposures under limits, and how to maximize a healthy working environment.
- For all personnel the rule is to keep exposures ALARA: i.e., keep exposures As Low As Reasonably Achievable.
- ALARA means that you must learn how to professionally execute your duties while responsibly managing yours and minimizing others' radiation exposures.

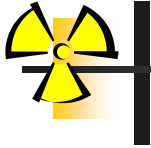
Remember These Key Points To Keep Exposure To Everyone at a Minimum



■ Time

- Physicians operating x-ray (fluoroscopy) equipment should keep the x-ray “beam-on time” to the minimum necessary. This saves dose to patients and to personnel.
- Personnel should limit the time they spend in unshielded environments to that which is necessary to complete their job properly.





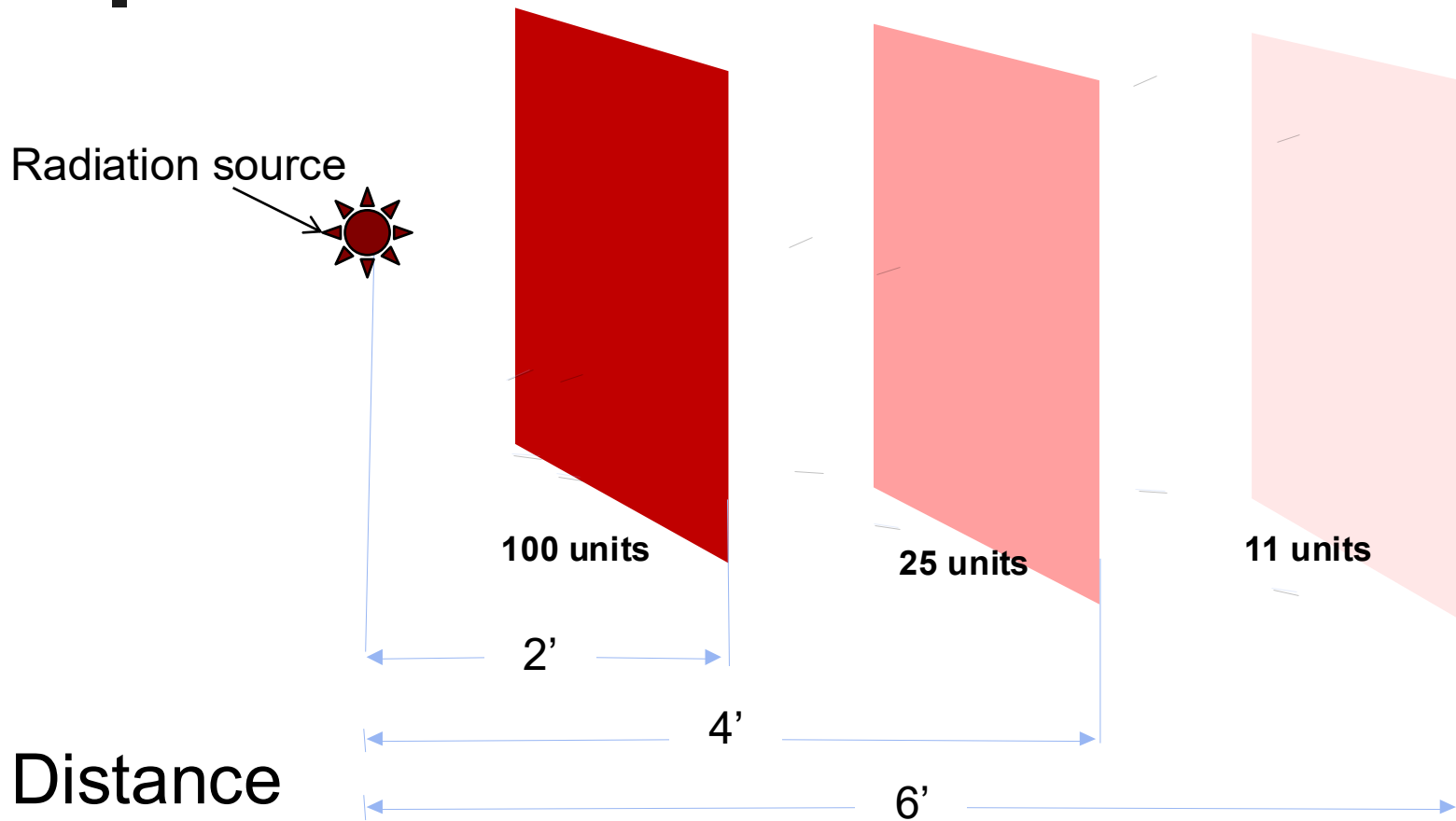
Minimizing Radiation Exposure

■ Distance

- To the extent consistent with appropriate medical care, maximize the distance between you and the area of the patient that is actively being x rayed.

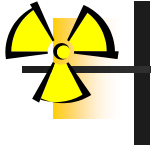


Minimizing Radiation Exposure

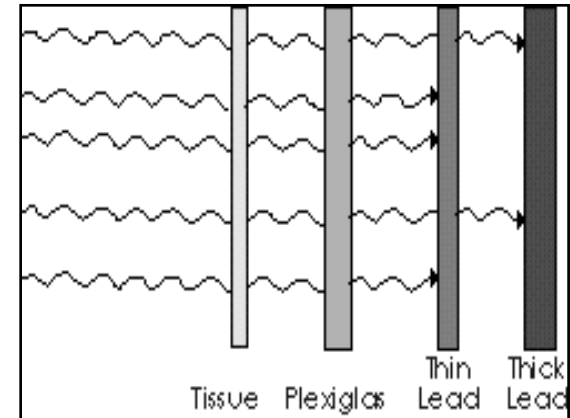


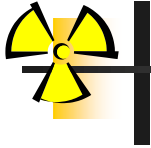
How distance from source corresponds to drop off in radiation exposure intensity

Shielding



- X rays are easily shielded by a thin layer of dense material, like lead.
- Protective aprons and leaded glass barriers in diagnostic radiology block most ($\sim 90\%$ - 99%) of the radiation.
- When working in fluoroscopy environments without a stand-alone radiation barrier for protection, personnel must wear a lead apron.





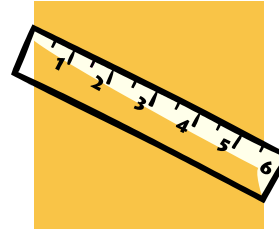
Minimizing Exposure

- Combining the use of

- Least Time



- Greatest distance



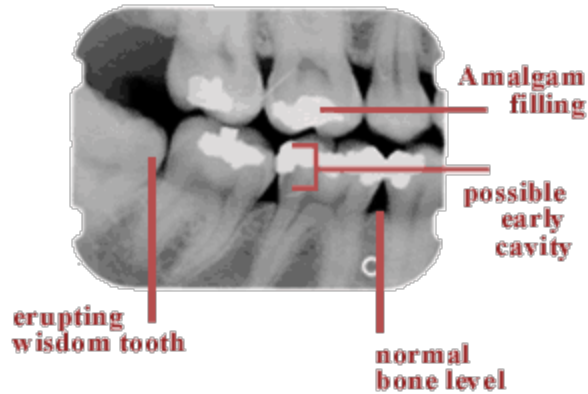
- Good shielding



= ALARA

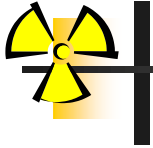
X-Rays

Medical Applications



X-rays are used in medicine for medical analysis. Dentists use them to find complications, cavities and impacted teeth. Soft body tissue are transparent to the waves. Bones and teeth block the rays and show up as white on the black background





Below are some x-rays showing objects which have been swallowed by people. The examples show an open safety pin and a child's stomach with all the pieces of a magnetic toy re-aligned after he's swallowed them...one by one!





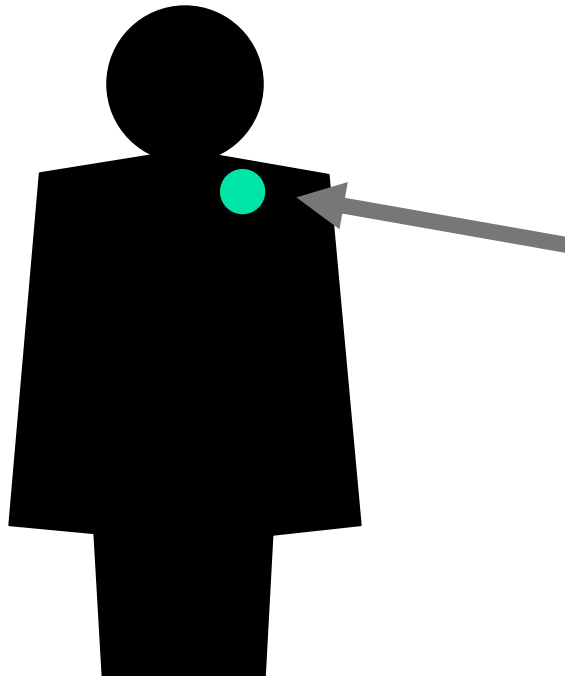
Personnel Monitoring

- Radiation dosimetry (e.g., whole body badge and/or finger ring) is provided to individuals who are **likely to receive an exposure greater than 10% of the annual limit**.
- The badge results are reviewed by Radiation Safety to ensure that radiation exposures are as low as reasonably achievable (ALARA).





Personnel Monitoring



Where should you wear your
collar badge?

At the collar
&
outside a lead apron

If you are enrolled in the personnel monitoring program, you must:

1. Wear your collar badge near the neck outside any protective apron
2. Wear any specially assigned abdomen badge at the waist under any protective apron
3. Store your badge away from radiation sources when not in use.



X-ray device – Radiographic Table



This picture X-ray tube in a collimated lead housing. The X-ray beam is pointed down to the table. The table is where the patient is placed and contains a slot for an X-ray film.



This is the mobile shield for operator. It is designed to protect operator from scattered X-rays (primarily from patient).

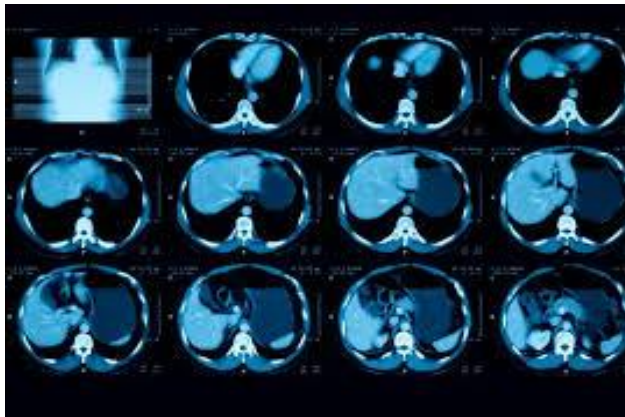


This is the control panel. Operator can select X-ray ON (exposure) time in fraction of minutes, the energy of X-ray (in kVp) and current applied (higher current = more X-rays).

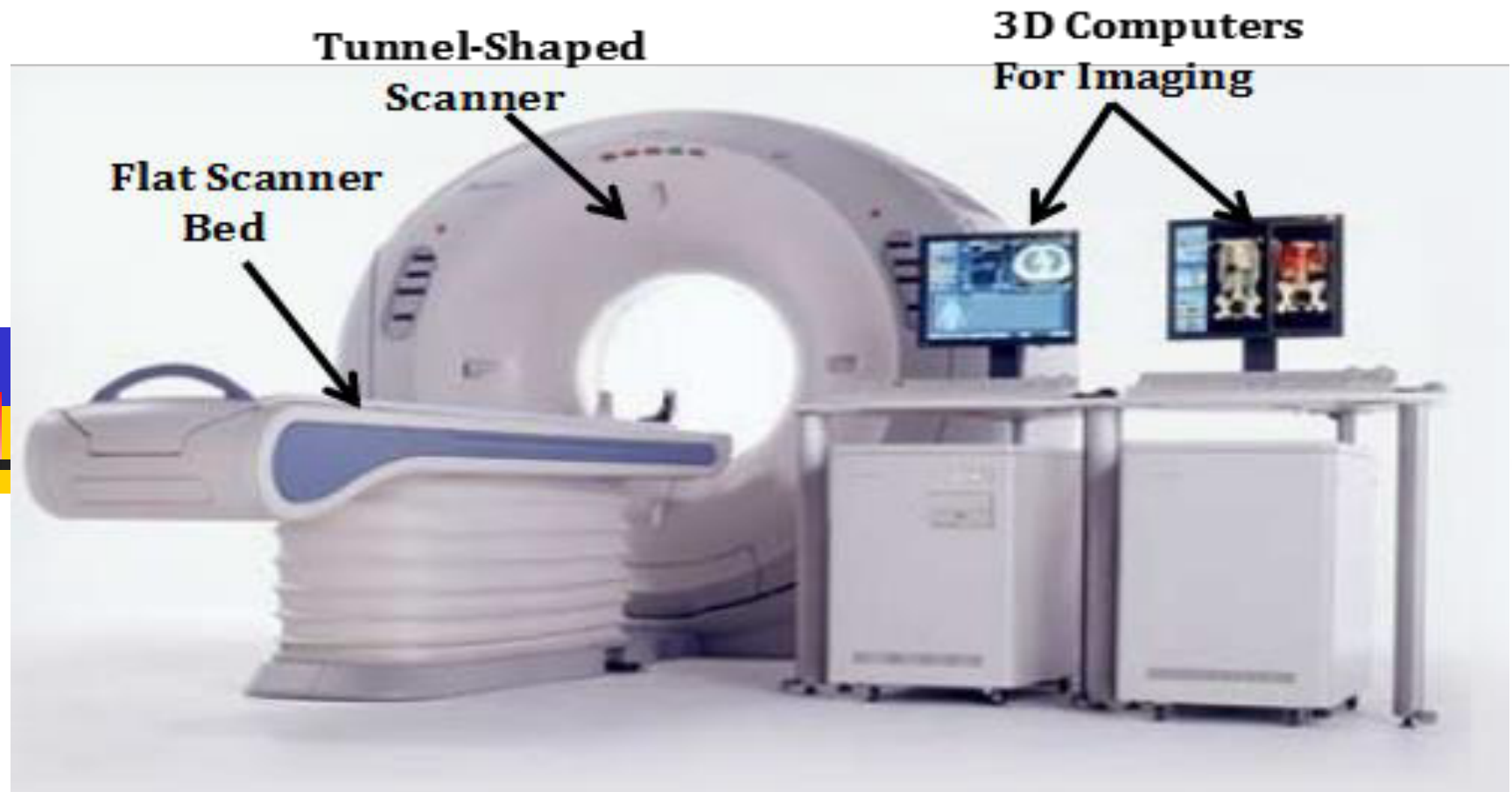


Title of Lecture: CT-Scan Lab.II

Hawler Medical University
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CT – Scan Procedure



**What are they, how do they work,
and important information for
successful implementation**

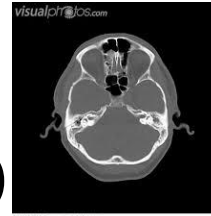


X-rays

Radiography

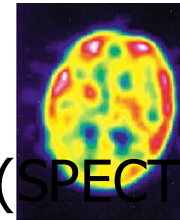


Computed Tomography (CT)



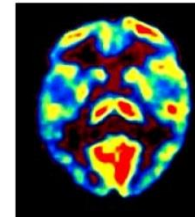
γ -rays (nuclear medicine)

Single Photon Emission Tomography (SPECT)

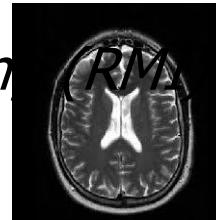


Positron Emission
Tomography (PET)

Non-ionizing radiation techniques



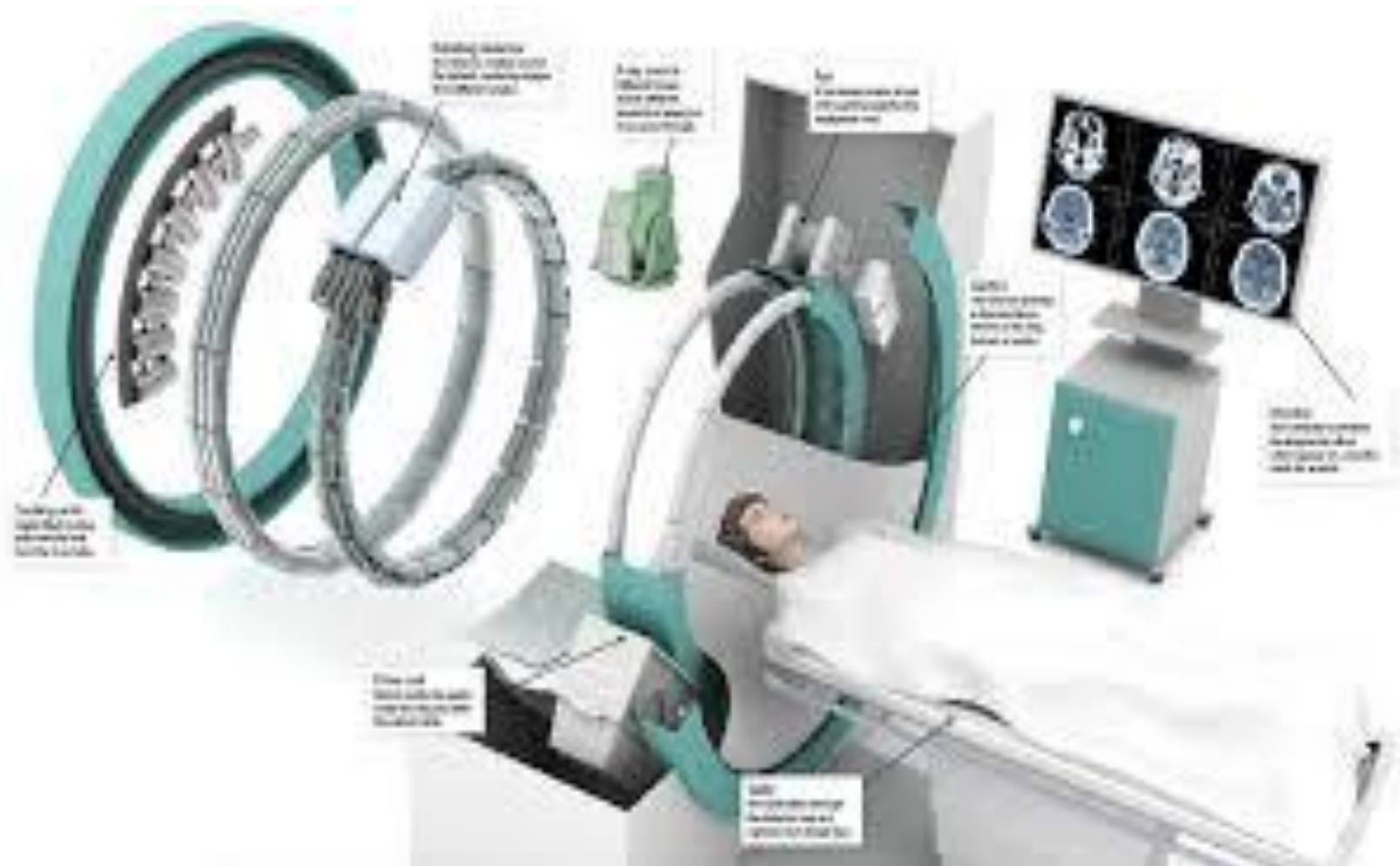
Magnetic Resonance Imaging (MRI)



Echography



Medipix detectors





Dose Notification Pop-up Window Examples

DOSE NOTIFICATION

One or more group result in a projected dose exceeding the Notification Value set. Select Cancel to go back to Viewedit and adjust scan parameters if clinically appropriate to set below the Notification Value. Selecting Confirm will proceed to scan and log user confirmation of scan parameters exceeding the Notification Value.

Series#	1	Series Description	A/P Onocology
Images	NV	Projected	
CTDIvol (mGy)	1 - 9	30	31.09

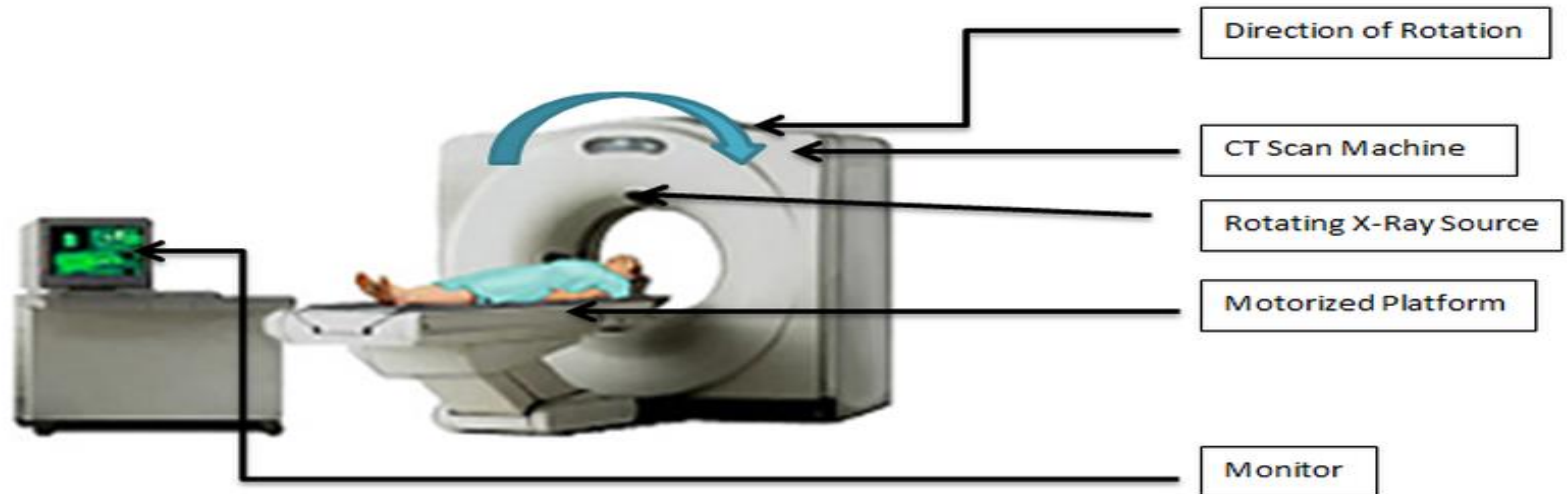
Diagnostic Reason

Confirm Cancel



Log files may include

- time
- date
- patient ID
- study ID
- event type
- series description
- operator name
- operator-provided reason for event
- did operator use password
- CTDIvol value that triggered notification event
- programmed notification value (CTDIvol)
- DLP value that triggered notification event
- programmed notification value (DLP)



Thank you

